
LECTURE NOTES

Discrete Mathematics notes

Proposition and logic

DM549 - Discrete Mathematics

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Fall 2024



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1 Propositions

Definition: Proposition

A proposition is a statement that is either true or false.

Example: Propositions

- $2 + 2 = 4$ is a proposition, and it is **true**.
- $x + 2 = 1$ is **not** a proposition, since its truth value depends on x .
- $2 + 2 = 5$ is a proposition, and it is **false**.

Propositions are often abbreviated as T (True) or F (False).

Notation: True and false

In addition to T and F, the symbols $\top$ (top) and $\perp$ (bottom) are sometimes used to denote **true** and **false** respectively. Both notations mean the same thing and may be used interchangeably throughout these notes.

2 Logical operators

Logical operators combine propositions into compound propositions, allowing us to express more complex statements concisely.

The most important operators are:

- Negation: $\neg$ (not)
- Conjunction: $\wedge$ (and)
- Disjunction: $\vee$ (or)
- Implication: $\Rightarrow$ (implies)
- Bi-implication: $\Leftrightarrow$
- Exclusive or: $\oplus$

Operator precedence (highest to lowest):

- 1) Negation: $\neg$ (not)
- 2) Conjunction: $\wedge$ (and)
- 3) Disjunction: $\vee$ (or)
- 4) Implication: $\Rightarrow$ (implies)
- 5) Bi-implication: $\Leftrightarrow$

There is no universally agreed-upon precedence for exclusive or ($\oplus$).

3 Truth tables

Truth tables define the operators exhaustively by listing every combination of truth values for p, q etc. .

Truth tables in discrete mathematics have a number of rows equal to 2^n , where n is the number of variables.

Example: number of rows

Given the proposition $p \vee q \wedge r$

Any truth table for this proposition must have $2^3 = 8$ rows

p	q	r
$\top$	$\top$	$\top$
$\top$	$\top$	$\perp$
$\top$	$\perp$	$\top$
$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\top$
$\perp$	$\top$	$\perp$
$\perp$	$\perp$	$\top$
$\perp$	$\perp$	$\perp$

3.1 Conjunction

True only if both propositions are true.

p	q	$p \wedge q$
$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\perp$
$\perp$	$\perp$	$\perp$

3.2 Negation

The opposite of a proposition p .

p	$\neg p$
$\top$	$\perp$
$\perp$	$\top$

3.3 Disjunction

True if at least one of the propositions is true.

p	q	$p \vee q$
$\top$	$\top$	$\top$
$\top$	$\perp$	$\top$
$\perp$	$\top$	$\top$
$\perp$	$\perp$	$\perp$

3.4 Implication

False only when p is true and q is false.

p	q	$p \rightarrow q$
$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\top$
$\perp$	$\perp$	$\top$

3.5 Bi-implication

True only if both propositions have the same truth value.

p	q	$p \iff q$
$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\perp$
$\perp$	$\perp$	$\top$

3.6 Exclusive or

True only if the propositions have opposite truth values.

p	q	$p \oplus q$
$\top$	$\top$	$\perp$
$\top$	$\perp$	$\top$
$\perp$	$\top$	$\top$
$\perp$	$\perp$	$\perp$

4 Tautology, contradiction, and contingency

Definition: Tautology

A proposition that is always true, no matter the truth values of its variables, is called a **tautology**.

Example: A tautology

We want to show that $p \vee (q \rightarrow \neg(p \wedge q)) \equiv \neg(q \wedge \neg p)$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$q \rightarrow \neg(p \wedge q)$	$p \vee (q \rightarrow \neg(p \wedge q))$	$\neg p$	$q \wedge \neg p$
$\top$	$\top$	$\top$	$\perp$	$\perp$	$\top$	$\perp$	$\perp$
$\top$	$\perp$	$\perp$	$\top$	$\top$	$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\perp$	$\top$	$\top$	$\top$	$\top$	$\top$
$\perp$	$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$	$\perp$

Comparing the two sides shows that $p \vee (q \rightarrow \neg(p \wedge q))$ is **always true** — it is a tautology.

Definition: Contradiction

A proposition that is always false, no matter the truth values of its variables, is called a **contradiction**. A contradiction is the opposite of a tautology.

Example: A contradiction

$(p \wedge q) \wedge \neg(p \vee q)$ is a contradiction:

p	q	$(p \wedge q) \wedge \neg(p \vee q)$
$\top$	$\top$	$\perp$
$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\perp$
$\perp$	$\perp$	$\perp$

Definition: Contingency

A proposition that is neither a tautology nor a contradiction — i.e. it is true for some truth assignments and false for others — is called a **contingency**.

Example: A contingency

$p \wedge q \iff \neg q$ is a contingency:

p	q	$p \wedge q$	$p \wedge q \iff \neg q$
T	T	T	⊥
T	⊥	⊥	T
⊥	T	⊥	⊥
⊥	⊥	⊥	T

Since the result is a mix of T and F, $p \wedge q \iff \neg q$ is a contingency.

5 Logical equivalence

Two propositions are **logically equivalent** if they have the same truth table. Tautology and logical equivalence are closely related, and equivalence is written with $\equiv$.

Definition 1.3.2

We call two propositions s, t **logically equivalent**, written $s \equiv t$, if $s \iff t$ is a tautology.

Note:

- In other words, s and t are two ways of saying the same thing.
- To find out whether $s \equiv t$, instead of constructing the truth table for $s \iff t$, one can compare the truth tables for s and t .
- The symbol $\equiv$ is not a logical operator, so $s \equiv t$ is not considered a compound proposition (while $s \iff t$ is).

Example: $p \Rightarrow q$ teq not p or q

p	q	$p \rightarrow q$	$\neg p \vee q$
$\top$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\perp$
$\perp$	$\top$	$\top$	$\top$
$\perp$	$\perp$	$\top$	$\top$

The two columns match for every row, so $p \rightarrow q \equiv \neg p \vee q$.

There are many ways to find equivalences. A few useful ones are covered below, starting with the distributive laws.

Distributive laws (Example 1.3.4)

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Table 5: p or $(q$ and $r)$ and $(p$ or $q)$ and $(p$ or $r)$ are logically equivalent

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T
T	T	⊥	⊥	T	T
T	⊥	T	⊥	T	T
T	⊥	⊥	⊥	T	T
⊥	T	T	T	T	T
⊥	T	⊥	⊥	⊥	⊥
⊥	⊥	T	⊥	⊥	⊥
⊥	⊥	⊥	⊥	⊥	⊥

The distributive law only works with Or and And — the method is to build the truth table for both sides and compare the whole way through.

5.1 Truth tables for equivalence and contraposition

Contraposition (Table 1.3.7, line 2)

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Intuition:

- If q is F, then $p \rightarrow q$ only becomes T if p is F.
- This is what $\neg q \rightarrow \neg p$ states.

Proof:

p	q	$p \rightarrow q$	$\neg q$	$\neg p$
$\neg q \rightarrow \neg p$	T	T	T	⊥
⊥	T	T	⊥	⊥
T	⊥	⊥	⊥	T
T	⊥	T	T	⊥
⊥	T	T	T	T

Equivalences involving implications (2): formulation using and , or , not (Table 1.3.7, line 1)

$$p \rightarrow q \equiv \neg p \vee q$$

Intuition:

- If p is F, both propositions are T.
- If p is T, for either proposition to be T, q must be T.

Proof:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	⊥	T
T	⊥	⊥	⊥	⊥
⊥	T	T	T	T
⊥	⊥	T	T	T

Equivalences involving implications (3): the implication and the bi-implication (Table 1.3.8, line 1)

$$(p \rightarrow q) \wedge (q \rightarrow p) \equiv p \Leftrightarrow q$$

Intuition: The rewritten left-hand side $(p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$ means both p and q need to have the same truth value.

Proof:

p	q	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$(p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$	$p \Leftrightarrow q$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\top$	$\perp$	$\perp$	$\perp$
$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$

Note: This justifies the notation of $\Leftrightarrow$ and saying “ p if and only if q ”.

5.2 Sets of numbers

Definition: Some sets of numbers

Important sets of numbers used in this and later lectures:

- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is the set of **integers**.
- $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ is the set of **positive integers**.
- $\mathbb{Z}^- = \{\dots, -3, -2, -1\}$ is the set of **negative integers**.
- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ is the set of **natural numbers**. In some sources this starts at 1.
- $\mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\right\}$ is the set of **rational numbers**, i.e. numbers given by any non-empty finite sequence of digits before the comma (possibly just 0) and any sequence of digits after the comma (possibly the empty sequence).
- $\mathbb{R}$ is the set of **real numbers**.
- $\emptyset$ is the **empty set** (contains no members).

5.3 Open propositional functions

Definition: Open propositional function

An open proposition (propositional function) is a statement that contains one (or more) variables, usually named something like $P(x)$.

Remarks:

- The variables usually represent numbers.
- When the variables are replaced with actual values, one obtains a proposition.
- For now, we will only focus on open propositions with a single variable.

Example: Evaluating a propositional function

We define a propositional function $P(x) \equiv 2x > x$.

$$P(-1) \equiv -2 > -1 \equiv \perp$$

$$P(0) \equiv 0 > 0 \equiv \perp$$

$$P(1) \equiv 2 > 1 \equiv \top$$

$$P(2) \equiv 4 > 2 \equiv \top$$

6 Quantifiers

Tips for quantifiers:

- There are three quantifiers: $\forall x$, $\exists x$, $\exists! x$.
- D stands for **domain**, i.e. the universe the quantifier ranges over.
- Quantifiers are placed before logical operators.
- The order of nested quantifiers matters quite a lot.

6.1 Universal quantifier

For a given propositional statement, one can say something about **all** its variables. The universal quantifier says something is true for **every** x in the domain, and is read as:

$$\forall x \in D : P(x)$$

“For all x in the domain D , $P(x)$ ”

Definition 1.4.1

For a propositional function $P(x)$, the statement

$$\forall x \in D : P(x)$$

is equivalent to the statement that $P(x)$ is true for all x in the set D . We call $\forall$ the **universal quantifier**.

Example: Universal quantifier

$$\forall x \in \mathbb{Z} : 2x > x,$$

so it matters what D is!

$$\forall x \in \emptyset : 2x > x \equiv \top$$

Let's say you have never driven a Ferrari. Then "whenever I have driven a Ferrari, I have crashed it" is **true**!

Comments:

- 1) All x belonging to the real numbers, so $2x > x$ (False).
- 2) All x belonging to the empty set, so $2x > x$ (True).

6.2 Existential quantifier

$$\exists x \in D : P(x)$$

is read as “there exists at least one x in the domain such that $P(x)$ is true”.

$$\exists x \in \emptyset : P(x) \equiv \perp$$

Definition 1.4.2

For a propositional function $P(x)$, the statement

$$\exists x \in D : P(x)$$

is equivalent to the statement that there exists at least one x in the set D such that $P(x)$ is true. We call $\exists$ the **existential quantifier**.

Remarks:

- Read: “there exists x in D such that $P(x)$ is true” or “for some x in D , $P(x)$ ”.
- An existential quantification over the empty set is always false.
- The existential quantification is true as long as there exists at least one x with the specified property, not just precisely one.

Example: Existential quantifier

Let $Q(x) \equiv x^2 = 4$.

Then $Q(x)$ can be unfolded as the (infinite) disjunction $Q(1) \vee Q(2) \vee Q(3) \vee \dots$, which is true as soon as **one** disjunct is true.

$$\exists x \in \mathbb{Z} : Q(x) \equiv \top$$

since $Q(2) \equiv 2^2 = 4 \equiv \top$ – we only need **one** x making the proposition true, not all of them.

6.3 Uniqueness quantifier

$$\exists! x \in D : P(x)$$

“There exists precisely one x in the domain such that $P(x)$ is true”

Definition: Uniqueness quantifier

For a propositional function $P(x)$, the statement

$$\exists! x \in D : P(x)$$

is equivalent to the statement that there exists precisely one x in the set D such that $P(x)$ is true. We sometimes call $\exists!$ the **uniqueness quantifier**.

Remarks:

- Read: “there exists precisely one x in D such that $P(x)$ (or “ $P(x)$ ” is true)”.

Example: Uniqueness quantifier

Let $Q(x) \equiv x^2 = 4$.

Over $D = \mathbb{Z}$, both $x = 2$ and $x = -2$ satisfy $Q(x)$, so there is **more than one** solution:

$$\exists! x \in \mathbb{Z} : Q(x) \equiv \perp$$

But over $D = \mathbb{Z}^+$, only $x = 2$ satisfies $Q(x)$:

$$\exists! x \in \mathbb{Z}^+ : Q(x) \equiv \top$$

6.4 Quantifiers with operators

(How do you read quantifiers?)

Method: when you get an assignment with quantifiers and operators, proceed as follows:

$$\forall x \in \mathbb{Z} : x \leq 3 \leftrightarrow 4 > 5 \equiv (\forall x \in \mathbb{Z} : x \leq 3) \wedge (4 > 5) \equiv \top$$

- 1) **Read the domain and the quantifier.** “For all x ’s that belong to the integers, so $x \leq 3 \leftrightarrow 4 > 5$, and true.”

Now we only have one $\cdot \leftrightarrow \cdot$ quantifier, so first, what does the sign say when isolated?

For all x ’s to belong to $\mathbb{Z}$ (the integers), so it must hold that $x \leq 3$. Here is $x \leq 3$ the part inside $\mathbb{Z}$ (integers), so the whole thing either holds or does not.

- 2) **Take the proposition and fill in the numbers** — first 3. You can then ask yourself “For all x ’s that belong to $\mathbb{Z}$, so it is true that $x \leq 3$. Does that hold or not?”
- 3) **Many people stop here for the next proposition.** The bar is just “ $4 > 5$ ” which is obviously false.
- 4) **Look at the operator from the Truth Table Bi-implication.** iff both propositions have the same truth value. Since neither proposition holds, the whole thing is true.

6.5 Restricted domains in quantifiers

This is about restricting the domain, so you add x (as input) belonging to D , telling something on the other side of the propositional function.

Definition: Restricted domains

For $x \in D, Q(x): P(x) \equiv \forall x \in D : (Q(x) \rightarrow P(x))$

Meaning: for all x 's that belong in $Q(x)$, so is it the case that $Q(x)$ implies $P(x)$ (because $P(x)$ and logically equivalent with $Q(x)$)

- $\forall x \in D, Q(x) : P(x) \equiv \forall x \in D : (Q(x) \rightarrow P(x))$,
- $\exists x \in D, Q(x) : P(x) \equiv \exists x \in D : (Q(x) \wedge P(x))$,
- $\exists! x \in D, Q(x) : P(x) \equiv \exists! x \in D : (Q(x) \wedge P(x))$.

Example: Restricted domains

Let $Q(x) \equiv 2x > x + 4$.

Then $Q(5) \wedge Q(6) \wedge Q(7) \wedge \dots \equiv \top$

$$\equiv \forall x \in \{5, 6, 7, \dots\} : Q(x)$$

$$\equiv \forall x \in \mathbb{Z}, x \geq 5 : Q(x)$$

$$\equiv \forall x \in \mathbb{Z} : (x > 5 \rightarrow Q(x))$$

$$\exists x \in \mathbb{Z}, x \leq 4 : Q(x) \equiv \exists x \in \mathbb{Z} : (x \leq 4 \wedge Q(x))$$

6.6 Nested Quantifiers with multiple variables

One can use multiple quantifiers when having more variables

7 Good to know equivalences

Table 6: Logical equivalences

Equivalence	Name
$p \wedge \top \equiv p$ $p \vee \perp \equiv p$	Identity laws
$p \vee \top \equiv \top$ $p \wedge \perp \equiv \perp$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \top$ $p \wedge \neg p \equiv \perp$	Negation laws

Table 7: Logical equivalences involving conditional statements

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

Table 8: Logical equivalences involving biconditional statements

$$p \iff q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \iff q \equiv \neg p \iff \neg q$$

$$p \iff q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \iff q) \equiv p \iff \neg q$$

8 Satisfiability

1.3.5 Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true (that is, when it is a tautology or a contingency). When no such assignment exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**. Note that a compound proposition is unsatisfiable if and only if its negation is true for all assignments of truth values to the variables, that is, if and only if its negation is a tautology.

When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a **solution** of this particular satisfiability problem. However, to show that a compound proposition is unsatisfiable, we need to show that **every** assignment of truth values to its variables makes it false. Although we can always use a truth table to determine whether a compound proposition is satisfiable, it is often more efficient not to, as Example 9 demonstrates.

Some exercises will ask you to determine the truth value from a **given** set of values for a proposition. Remember that talking about satisfiability really just means asking: “Can this statement be true?” If there is a way to make a statement true, we say that it is **satisfiable**.

Example: Determining satisfiability from given values

(a) If p is true and q is false, then $p \wedge q$ is false.

Even if it wasn't stated on the slides, we can use the satisfiability argument: we simply assume the truth values of p and q and evaluate the operator.

$$p = \top, q = \perp$$

$$p \wedge q = \top \wedge \perp = \perp$$

From the question we were asked, we can conclude that this is correct.

Recall from the CS (Datalogi) course:

Satisfiability (Boolean expressions)

A Boolean expression with n variables $x_1, x_2, \dots, x_n$ is called **satisfiable** if there exists a set of values (e.g. $x_1 = \perp, x_2 = \top, \dots, x_n = \perp$) that makes the whole expression true.

I.e. if there exists a row in the expression's truth table where the output is 1.

The SAT problem: given a Boolean expression with n variables, decide whether it is satisfiable (and if so, find a satisfying set of values for $x_1, x_2, \dots, x_n$).